

Review

Blockchain-Based Fraud Detection: A Comparative Systematic Literature Review of Federated Learning and Machine Learning Approaches

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Abstract

This systematic literature review uses the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology to assess progress in blockchain-based Federated learning (FL) and Machine Learning (ML) for detecting financial fraud over the last five years (2020–2025). An initial pool of 29,274 records identified across IEEE Xplore, ACM Digital Library, and ScienceDirect yielded 1585 peer-reviewed studies that met the inclusion criteria. Both qualitative and quantitative approaches were used. The examined papers were classified according to algorithm type, fraud types, and evaluation measures. Credit card fraud and cryptocurrency fraud dominated the literature, with supervised learning (e.g., XGBoost, 95% accuracy) and federated learning (e.g., FedAvg, 91% accuracy) emerging as dominant methodologies. Centralized ML outperforms FL in latency but poses privacy risks. FL–blockchain hybrids reduce false positives. While precision, recall, and F1-score are commonly used, few studies use cost-sensitive criteria. Future research should prioritize adaptive FL aggregation, privacy-preserving ML, and cross-industry collaboration.

Keywords: blockchain; federated learning; fraud detection; machine learning; privacy-preserving AI; financial security



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1. Introduction

Financial fraud is a major issue in industries such as banking, insurance, and healthcare, resulting in enormous economic losses and societal harm [1,2]. Traditional detection approaches, which rely on human processes, are inefficient, expensive, and prone to errors [3,4]. As the number of digital transactions increases, fraudsters use more complex strategies necessitating better detection mechanisms [5,6].

Artificial intelligence (AI) and machine learning (ML) have developed as effective fraud detection methods, utilizing big datasets to discover aberrant patterns [7,8].

Supervised and unsupervised machine learning approaches, including classification algorithms, are frequently employed [6,9]. However, centralized ML techniques face

several issues, including data privacy concerns, single points of failure, and vulnerability to adversarial attacks [10–16].

Blockchain technology provides a decentralized, transparent, and immutable framework for fraud detection systems, increasing their security and trustworthiness [17–20]. Coupled with federated learning (FL), a privacy-preserving ML paradigm in which models are trained cooperatively without disclosing raw data [21,22] addresses significant drawbacks of centralized systems [23–26]. FL addresses privacy problems under legislation such as GDPR [27], whilst blockchain ensures auditability and resilience to tampering [28,29]. Despite increased interest, recent studies have mainly focused on specific sectors such as credit card fraud [30,31], healthcare fraud [32], and payment fraud [33], with no systematic comparison of blockchain–FL frameworks to standard ML. Recent research has highlighted FL’s potential for fraud detection [34,35] and blockchain’s role in securing FL [18,36,37] but none have thoroughly evaluated their combined efficacy for financial fraud [38]. Recent architectural work proves a unified blockchain-enabled FL framework is capable of cross-silo collaboration and secure update verification, underscoring the feasibility of such integration in financial domains [39]. However, recent emphasis on the emerging need for comparative frameworks and privacy-preserving mechanisms in federated learning for financial fraud detection stresses the same research gap addressed here [39].

This systematic literature review (SLR) closes the gap by identifying blockchain-based fraud detection systems with ML/FL integrations [20,24,40,41]; comparing the performance, privacy, and scalability of FL and standard ML for fraud detection [34,42,43]; evaluating datasets (e.g., Fed-Fraud [44]) and metrics for blockchain–FL research [26,36]; highlighting research gaps, including adversarial attacks [11,45–51] and regulatory compliance [27]; and comparing decentralized and centralized ML techniques [24,26,41,52]. Prior reviews emphasized components of FL, blockchain, or IoT security but rarely provided a systematic comparison of ML against FL for financial fraud or synthesized differential privacy constraints specific to finance [53,54]. The Study Selection Process is presented as a PRISMA flow diagram in Figure 1. The search strategy commenced with an extensive search across three major academic databases, IEEE Xplore, ACM Digital Library, and ScienceDirect, yielding a total of 29,274 records. No additional records were identified through registers or other sources. After exclusions by time period ($n = 19,646$), language ($n = 6745$), and publication type ($n = 1298$), 1585 studies were included in the final review.

The rest of this paper is organized as follows: Section 2 describes the research methodology, including the search criteria, study selection, data extraction, and quality evaluation. The SLR findings and the responses to the study questions are subsequently presented. The results and possible challenges that undermined the validity of this review are addressed, respectively. Finally, we provide a discussion and a conclusion of the study in the last section. This review helps researchers choose approaches for secure, scalable fraud detection. In this section, we present the results of the 1585 studies that met our inclusion and quality assessment criteria.

We included only those studies that satisfied our rigorous selection standards and were not flagged for bias during the screening process. Once the studies passed our inclusion criteria and underwent thorough coding and analysis, we did not conduct further bias assessments on their internal findings. Previous reviews often excluded recent conference papers and emerging empirical studies. Our analysis was based on quantitative descriptive summaries derived from these high-quality studies.

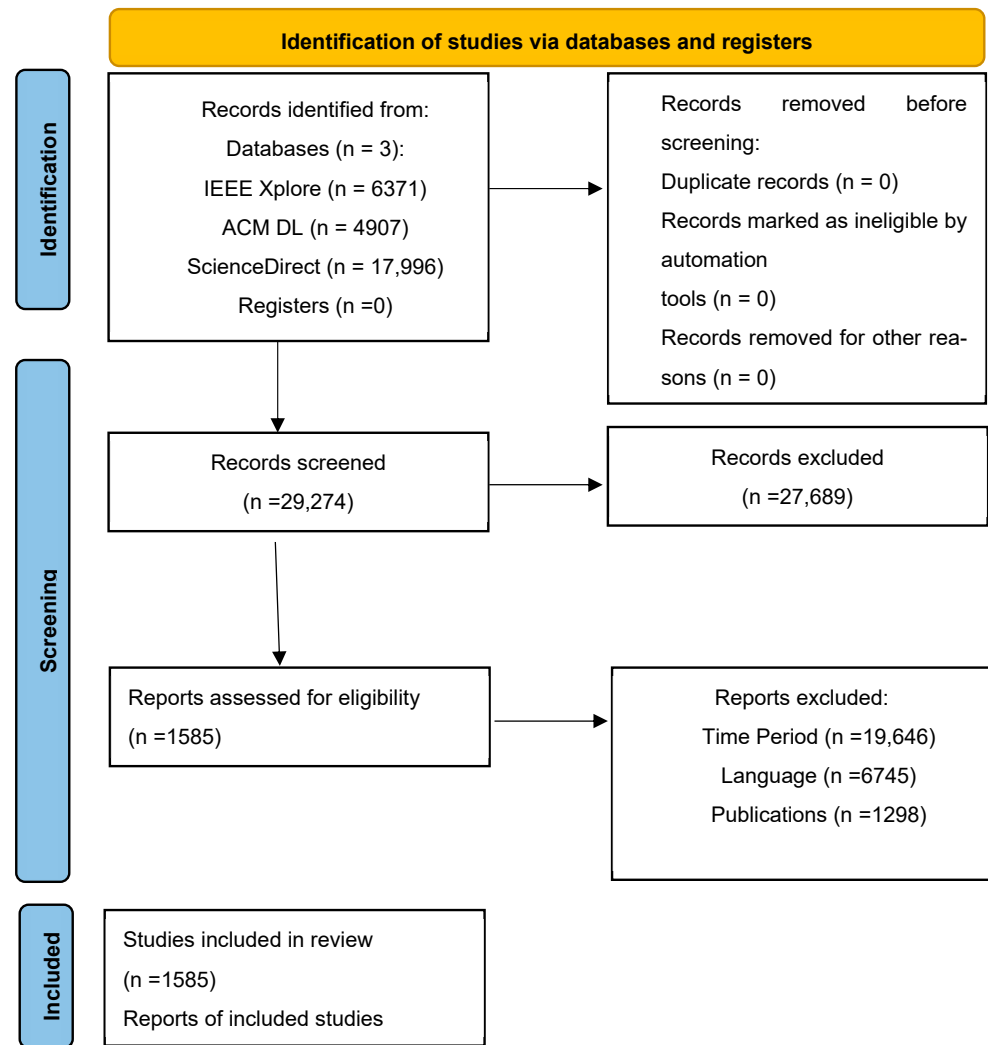


Figure 1. PRISMA flow diagram of search strategy.

2. Materials and Methods

With financial services going digital, there is a lot more data out there for payment fraud, credit card fraud, scams, identity theft, and money laundering, etc., despite the ability for fraud detection with ML and blockchain technology. Blockchain's secure ledgers and federated learning could help train models together without exposing private information [55,56]. However, these technologies remain under integrated in practice. Other options like basic rule-based systems and centralized ML make things worse, with a lot of false alarms, slow detection, and easy data hacks. So, this review compares machine learning and federated learning in the blockchain world, bridging the gap between theoretical advances and real-world implementation. The PICOC (Population Intervention Comparison Outcome Context) framework is written in Table 1.

Table 1. PICOC framework.

Element	Description	Examples
Population	Application area where anomaly detection is applied	Financial transactions, credit card transactions, banking data
Intervention	Methodology/technology used	Machine learning, federated learning, blockchain
Comparison	Alternative methods compared against current ones	Supervised learning, centralized ml, blockchain-based methods
Outcome	Results of the intervention	Improved accuracy, reduced false positives, real-time detection
Context	Setting where applied	Financial institutions, banking systems, e-commerce platforms

Research String:

("Blockchain")

AND

("Machine Learning" OR "Federated Learning" OR "Financial Transactions"

OR "Fraud Detection")

The inclusion and exclusion criteria are written in Table 2.

Table 2. Exclusion and inclusion criteria.

	Exclusion	Inclusion
1	Articles that do not focus on financial fraudulent transactions.	Articles that were conducted from 2020 to 2025.
2	Articles that do not concern the use of machine learning methods/blockchain technology/federated learning	Articles that focus on financial fraud detection and ML methods based on blockchain technologies and federated learning.
3	Articles that are in the form of abstracts, short papers, newspapers, posters, and book chapters.	A peer- reviewed research article or systematic literature review
4	Studies that were not published in the English language.	Studies were conducted in English only.
5	Studies that do not mention their performance evaluation metrics	Only review journal or conference papers related to financial fraud detection.

Quality Assessment Questions (QA)

(Each QA item was scored 0–2 (low to high) and the total score determined inclusion quality)

QA1: Is the domain of the study related to financial fraud detection? (2)

QA2: Does the study apply machine learning, federated learning, or blockchain for fraud detection? (2)

QA3: Are evaluation metrics such as accuracy, precision, recall, and F1-score or AUC clearly reported? (2)

Research Questions:

RQ1: What are the machine learning techniques employed for financial fraud detection employed in the literature?

RQ2: What are the evaluation metrics employed to detect financial fraud?

RQ3: What are the different types of financial fraudulent transactions covered in the literature review?

RQ4: How do blockchain-based federated learning and machine learning approaches differ in detecting financial fraud?

RQ5: What are the machine learning techniques employed for financial fraud detection employed in the literature?

ML has significantly changed how we catch money fraud. Each method is good at spotting different kinds of fake transactions. We can sort these methods into supervised, unsupervised, and hybrid types. We will look at how they perform, how they work, and how they are used in real life to stop fraud. By investigating these methods, we want to make it easier to understand what they are good at and what they are not good at when it comes to protecting money from new fraud tricks.

The extracted machine learning methods used across the selected studies are listed in Table 3, providing insight into how financial fraud detection has been approached in the literature.

Table 3. Machine learning techniques employed for financial fraud detection as identified in the literature.

References	Algorithm	Type (FL/ML)	Reported Results	Description
[22,57,58]	FedAvg Federated Averaging	Federated Learning	85–92% accuracy, 0.89 F1-score	Aggregates local model updates from distributed nodes while preserving data privacy x
[4,10,31,59,60]	Support Vector Machine (SVM)	Machine Learning	94% precision, 0.91 recall	Uses hyperplanes to classify data and is effective for high dimensional spaces
[17,61–63]	Random Forest	Machine Learning	AUC-ROC 0.95–0.97 86–94% accuracy	Collective of D.Tree with feature bagging/strong to overfitting
[32,64,65]	LSTM Autoencoders	Federated Learning	88% anomaly detection rate	Neural networks for sequential data anomaly detection in decentralized settings
[6,57,59,62,66]	XGBoost	Machine Learning	95% accuracy, 0.93 F1-score	Gradient-boosted trees with regularization to prevent overfitting and handles class imbalance
[67–69]	Federated XGBoost	Federated Learning	90% accuracy, 0.88 AUC	Distributed version of XGboost with secure aggregation
[70–72]	Graphical Neural Networks	Machine Learning	89% fraud recall	Analyzes transaction graphs to detect suspicious network patterns
[73–76]	FedProx	Federated Learning	91% accuracy (handles non-IID data)	FL variant with proximal term to address data heterogeneity
[27,77–79]	Differential Privacy FL	Federated Learning	87% accuracy ($\epsilon = 2.0$)	Adds noise to model updates to assure privacy
[80–82]	Q-Learning (RL)	Federated Learning	93% adaptive fraud detection	Reinforcement learning for dynamic policy updates in decentralized systems
[17,22]	Decision Tree (DT)	Machine Learning	89–92% accuracy	Interpretable but inclined to overfitting
[3,17]	Logistic Regression (LR)	Machine Learning	85–88% AUC-ROC	Predicts binary outcomes via logistic function, simple but limited to linear patterns and mostly used for financial mismanagement

Table 3. Cont.

References	Algorithm	Type (FL/ML)	Reported Results	Description
[5,83]	Deep Neural Networks (DNNs)	Machine Learning	90–93% F1-score	Multi-layer perceptrons for complex pattern detection; needs large data
[1,64]	Autoencoders	Machine Learning/Federated Learning	88–91% anomaly detection accuracy	Unsupervised NN for reconstruction error-based anomaly detection; learns normal patterns, detects deviations
[14,84–86]	Isolation Forest (IF)	Machine Learning	89% detection rate ($\uparrow$ speed vs. OC-SVM)	Unsupervised plus isolates irregularities via random splits in feature space and works with high-dim data
[7,63,87]	K-Means Clustering	Machine Learning	77–85% cluster purity	Groups similar transactions, e.g., money laundering and it groups similar transactions based on feature similarity. Used to detect unusual money flow patterns.
[17,60]	AdaBoost	Machine Learning	93% accuracy (imbalanced data)	Ensemble method with adaptive boosting and focuses on misclassified samples
[6,17,62,66]	LightGBM	Machine Learning	94% AUC-ROC	Gradient-boosting framework optimized for speed and efficiency
[61,88]	SMOTE + RF	Machine Learning	91% recall (class-imbalanced data)	Combines synthetic minority and oversampling with Random Forest
[89–91]	Transformer Models	Machine Learning	92% F1-score (sequential fraud)	Self-attention mechanisms for contextual fraud design detection
[4,10]	Hybrid SVM-DT	Machine Learning	90% precision	Combines SVM's margin maximization with DT interpretability
[82,92]	Deep Reinforcement Learning	Federated Learning	95% adaptive accuracy	Learns fraud policies through reward-based systems in surroundings of FL
[93,94]	Homomorphic Encryption FL	Federated Learning	86% accuracy (privacy-preserving)	Enables computations on encrypted data for safe FL
[95,96]	Zero-Knowledge Proof FL	Federated Learning	88% verifiable accuracy	Validates model integrity without revealing raw data

Federated Averaging (FedAvg):

FedAvg enables decentralized collaborative training without data sharing [22,97]. Clients train locally and share updates, which are averaged into a global model. It achieves 91–92% accuracy and 0.89 F1 [22,56], proving effective in healthcare and finance [56,68]. However, it faces poisoning risks, straggler issues, and reduced accuracy with skewed data [73,74].

Support Vector Machine (SVM):

SVM constructs optimal hyperplanes for classification, excelling in high-dimensional fraud detection with 94% precision and 0.91 recall. Kernel functions enable non-linear mapping, boosting resilience [31]. It outperforms LR and DT in imbalanced datasets [17,60], though it needs careful tuning. Despite scalability limits, it remains favored in regulated industries due to interpretability and risk minimization [31,59].

Random Forest (RF):

RF gathers multiple decision trees, reducing overfitting and achieving AUC-ROC 0.95–0.97. It handles imbalanced and noisy data effectively and is widely applied in credit rating and anomaly detection. While computationally heavier than single trees, RF's robustness, parallelization, and adaptability ensure strong adoption in finance and cybersecurity [3,6,17].

LSTM Autoencoders:

These deep models combine autoencoder compression with temporal modeling, achieving 88% anomaly detection in federated setups [83]. They preserve privacy by only sharing parameters, which is useful in IoT and healthcare. Applied to sequential anomalies, they detect faults and abnormal signals [32]. Recent improvements include differential privacy and attention mechanisms [89,98].

XGBoost:

A gradient-boosted tree algorithm achieving 95% accuracy and 0.93 F1 in fraud detection. Its regularization, parallelism, and imbalance handling make it superior to simpler ML models [6]. However, it requires fine-tuning and is less interpretable [59]. XGBoost is widely applied in credit risk, fraud scoring, and anti-money laundering [66].

Federated XGBoost:

An FL variant using secure aggregation, achieving 90% accuracy and 0.88 AUC [68,69]. It preserves privacy while retaining strong predictive power compared to centralized XGBoost [6]. Cryptographic costs add complexity, but hybrid blockchain approaches improve security [99]. Studies show it outperforms FedAvg in tabular fraud tasks [66].

Graph Neural Networks (GNNs):

GNNs model fraud as transaction graphs, detecting organized fraud with 89% recall [24]. They propagate relational information, outperforming non-graph ML by 18–30% [59,70]. Despite challenges in scaling and adversarial attacks, attention-based GNNs (GAT) improve performance. Applications include cryptocurrency and DeFi fraud detection [72].

FedProx:

FedProx introduces a proximal term to stabilize training with non-IID data, achieving 91% accuracy. It outperforms FedAvg in heterogeneous cross-silo settings like healthcare [73]. By mitigating stragglers and integrating differential privacy, it improves FL robustness [62,74]. It is being increasingly applied to finance under data restriction challenges.

Differential Privacy FL:

DP-FL adds calibrated noise to protect updates, ensuring $\epsilon = 2.0$ privacy with 87% accuracy. It safeguards GDPR/HIPAA compliance in sensitive domains [27,77]. While

noise reduces accuracy (up to 15% vs. non-private FL), it is stronger than k-anonymity [100]. Ongoing work is combining it with secure multiparty computation for resilience [94].

Q-Learning (Reinforcement Learning):

A reinforcement learning method that adapts policies in real time, reaching 93% accuracy in FL settings. It handles shifting fraud strategies and non-stationary patterns [11]. Blockchain-enhanced versions ensure secure logging and integrate Deep Q-Networks [68,82]. Studies show 16–20% recall improvement over rule-based systems [11].

Decision Tree (DT):

DTs achieve 88–92% accuracy in fraud detection, offering transparency, which is critical in banking regulation. They are prone to overfitting unless pruned but remain efficient for credit scoring. Recent hybrids improve anomaly detection in imbalanced fraud datasets. Their interpretability ensures continued relevance in fintech [9,17,21].

Logistic Regression (LR):

LR achieves 86–88% AUC-ROC in fraud detection. It is fast, interpretable, and regulator-friendly but struggles with non-linear patterns. Regularization (L1/L2) improves performance, though ensembles outperform LR by 5–10% [30,101]. Federated LR extends its use to privacy-preserving cross-bank applications [56,69].

Deep Neural Networks (DNNs):

DNNs achieve 90–93% F1-scores, capturing complex fraud patterns in high-dimensional data. Applications include NLP-based fraud analysis and check forgery detection. However, they demand high computation and labeled data [5]. Federated DNNs reduce risks of gradient leakage while enabling collaboration.

Autoencoders:

Unsupervised models with 88–91% accuracy in anomaly detection. They learn normal transaction patterns and detect deviations. Hybrid autoencoders with attention or SHAP improve interpretability and robustness. Real-world applications report a 30% false positive reduction in payment monitoring [1].

Isolation Forest (IF):

An unsupervised method using random splits, achieving 89% detection [14,84]. It is efficient ($O(n)$) and scalable, outperforming One-Class SVM in speed [34]. Extended IF improves stability for categorical features [85]. Deployed in banking, it cuts fraud inquiry times by 40% [84].

K-Means Clustering:

Detects suspicious groups with 77–85% cluster purity. Useful where labeled data is scarce, especially for AML. Variants like X-Means improve performance though are sensitive to scaling and K value [63,86]. Secure multi-party K-Means supports collaborative fraud detection [68]. Real deployments reduce false positives by 25% [86].

AdaBoost:

An ensemble-boosting model achieving 93% accuracy on imbalanced fraud data [21,60]. It emphasizes hard-to-classify cases and preserves interpretability [21]. Federated AdaBoost secures collaboration by exchanging only classifier weights [102]. Recall improvements of 15–20% are reported for minority fraud classes [60].

LightGBM:

A gradient-boosting framework with 94% AUC-ROC. Optimized for speed and large datasets, it supports categorical features and GPU acceleration [6,17]. Federated LightGBM secures distributed training [66,69]. In real deployments, it reduces false declines by 15% and improves fraud detection by 30% [17].

SMOTE + Random Forest:

Hybrid method addressing class imbalance with 91% recall. Used in rare fraud and medical claim datasets. It reduces false negatives by 40% compared to under-sampling [89]. Federated variants preserve privacy while sharing synthetic data [62].

Transformer Models:

Achieve 92% F1-score in sequential fraud detection [91]. They capture contextual dependencies in transactions and support transfer learning (e.g., Fin BERT) [89]. High computational costs are mitigated by aggregation and gradient masking in FL [103]. Real-world deployments detect coordinated multi-merchant attacks effectively [91].

Hybrid SVM-DT:

Combines SVM's classification strength with DT interpretability, reaching 90% precision [10]. Effective in regulated credit scoring where explanations are required [5]. Federated implementations ensure decentralized learning [104]. Studies show an 18% reduction in unjustified rejections compared to standalone models [10].

Deep Reinforcement Learning (DRL):

Achieves 95% adaptive accuracy for fraud detection in FL. Excels in dynamic contexts like cryptocurrency fraud. Innovations include curiosity-driven exploration and blockchain-secured logging [11,82]. Applied in AML, it improves fraud response times by 30% [105].

Homomorphic Encryption FL:

Enables encrypted model training with 86% accuracy. Though 10–100× slower, hardware acceleration aids deployment in SWIFT monitoring [62]. EU GDPR drives adoption, keeping performance within 5% of plaintext models. Hybrid methods combine encryption with secure multi-party computation for efficiency [91,93].

Zero-Knowledge Proof FL:

Provides verifiable model integrity with 88% accuracy [96]. Useful in credit scoring and AML audits [106]. Blockchain-enabled ZKPs ensure audit trails and detect poisoning attacks [107]. They address data sovereignty concerns in cross-border fraud networks, enhancing trust [96]. With their own benefits, FL and ML techniques have shown great promise in blockchain-based fraud detection systems as in RQ1.

RQ2: What are the evaluation metrics employed to detect financial fraud?

In order for fraud detection algorithms to be effective, their performance must be rigorously evaluated using metrics that take into consideration the unbalanced nature of financial datasets, where illicit transactions make up a very small portion of total activity. In these situations, traditional accuracy metrics frequently turn out to be deceptive, leading researchers to use specialized metrics that more accurately reflect model precision, recall trade-offs, and economic consequences [15]. The main assessment frameworks used in the literature are examined in this part, along with their applicability to various fraud detection scenarios. These frameworks include precision-recall analysis, ROC-AUC, F1-score, and cost-sensitive metrics. We examine how these metrics tackle important issues such as unreported fraud cases and false positives in transaction monitoring, giving practitioners a methodological basis on which to choose the best validation techniques for their particular use cases. The evaluation metrics are written in Table 4.

Table 4. Evaluation metrics.

Ref	Metric	Formula	Interpretation	When to Use	Trade-Offs
[3,17,21,66]	Accuracy	$(TP + TN)/(TP + TN + FP + FN)$	Overall correctness of predictions	Balanced datasets	Misleading for imbalanced data (rare fraud)
[17,31,59,66,70]	Precision	$TP/(TP + FP)$	% of predicted frauds that are actual frauds	When FP costs are high (e.g., false alarms)	High precision $\rightarrow$ low recall
[6,108]	Recall (Sensitivity)	$TP/(TP + FN)$	% of actual frauds correctly detected	When FN costs are high (e.g., missed frauds)	High recall $\rightarrow$ low precision
[8,17,42,56,66,68]	F1-Score	$2 \times (\text{Precision} \times \text{Recall})/(\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall	Balanced view of both FP and FN	Favors models with similar precision/recall
[9,21,88]	Specificity	$TN/(TN + FP)$	% of legitimate transactions correctly identified	When FP reduction is critical	Less emphasized in fraud detection
[6,21,34,67,71,83,108]	AUC-ROC	Area under ROC curve	Model's ability to distinguish fraud vs. non-fraud across thresholds	Overall performance comparison	Insensitive to class imbalance
[3,109–111]	False Positive Rate (FPR)	$FP/(FP + TN)$	% of legitimate transactions flagged as fraud	Cost analysis of false alarms	Lower FPR $\rightarrow$ higher threshold $\rightarrow$ potential high FN
[30,40,101]	False Negative Rate (FNR)	$FN/(FN + TP)$	% of frauds missed by the model	Risk assessment of undetected fraud	Lower FNR $\rightarrow$ lower threshold $\rightarrow$ potential high FP
[33,43,84,87,109,110]	Precision-Recall Curve (AUC-PR)	Area under P-R curve	Performance under class imbalance (focus on fraud class)	Highly imbalanced datasets	More informative than ROC when fraud is rare
[21,58,112,113]	Matthews Correlation Coefficient (MCC)	$(TP \times TN - FP \times FN)/\sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN))}$	Balanced measure for binary classification	When all four confusion matrix categories are important	More reliable for imbalanced data than accuracy
[20,61,63,86,88]	G-Mean	$\sqrt{(\text{Recall} \times \text{Specificity})}$	Geometric mean of sensitivity and specificity	Imbalanced datasets where both classes matter	Less sensitive to majority class than accuracy
[3,53,114]	Balanced Accuracy	$(\text{Recall} + \text{Specificity})/2$	Average of recall and specificity	Imbalanced classification problems	Simpler alternative to G-Mean
[31,59,96]	Detection Prevalence	$(TP + FP)/\text{Total}$	Proportion of cases flagged as fraud	Understanding system's alerting behavior	Does not measure correctness, just volume of alerts
[6,56,115]	Positive Predictive Value (PPV)	$TP/(TP + FP)$	Same as precision	When focusing on predictive value of positive cases	Synonym for precision
[9,21,116]	Negative Predictive Value (NPV)	$TN/(TN + FN)$	% of predicted negatives that are actual negatives	When focusing on predictive value of negatives	Less commonly used in fraud detection
[84,87]	Youden's J Index	$\text{Recall} + \text{Specificity} - 1$	Combines sensitivity and specificity	Finding optimal threshold	Ranges from -1 to 1 (1 = perfect test)
[17,59,74,80]	Lift Score	$(TP/(TP + FP)) \div ((TP + FN)/\text{Total})$	How much better model performs than random models	Marketing and business applications	Depends on base rate of positive class
[3,114,117]	Cost-Sensitive Accuracy	$(w_1 \times TN + w_2 \times TP)/\text{Total}$	Accuracy with different weights for classes	When misclassification costs are unequal	Requires domain knowledge to set weights
[30,101,110]	Expected Cost	$C_{10} \times FP + C_{01} \times FN$	Total expected cost of misclassifications	When costs of FP and FN are known	Requires accurate cost estimates

In financial fraud detection, the evaluation metric is critical for assessing model performance [87,101,118,119].

However, no defined assessment measures are strictly employed to evaluate ML methods for fraud detection [101]. Different researchers have recently used a variety of performance evaluations as shown below:

Accuracy:

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$ measures overall correctness [3,17]. In fraud datasets with >99% legitimate cases, it can be misleading (e.g., 99% accuracy with 0% fraud detection) [21]. Studies show it overestimates performance by 16–30% compared to balanced metrics [17]. It should be reported only alongside class-sensitive metrics like F1 [3].

Precision:

Precision = $TP / (TP + FP)$, vital where false positives incur costs. Banking systems often prioritize >90% precision to maintain trust, though this lowers recall [71]. Precision is GDPR-relevant under “right to explanation”. Increasing thresholds (95% → 98%) may cut fraud detection by 40% [31,59].

Recall (Sensitivity):

Recall = $TP / (TP + FN)$, crucial where missed frauds are costly. Banks often tolerate more FPs to maximize recall, though this raises review workload [108]. In FL, recall variance highlights fraud heterogeneity (e.g., 15% gap between banks). High recall may increase costs by 25–50% [6,115].

F1-Score:

$F1 = 2 \cdot (\text{precision} \times \text{recall}) / (\text{precision} + \text{recall})$, balancing both [56,68]. It is more reliable than accuracy for imbalanced fraud data, with Kaggle winners prioritizing it 3:1 [42]. Yet, equal weighting may mask real-world trade-offs (e.g., 80/90 vs. 90/80 precision/recall yield the same $F1 = 0.85$) [68].

Specificity:

Specificity measures correct recognition of legitimate cases, often >99.9% in fraud systems [9,21]. Even 0.1% FP can cause thousands of alerts in high-volume systems [87]. Thresholds are adjusted seasonally (e.g., holiday shopping surges) to minimize customer disruption [21].

AUC-ROC:

AUC-ROC measures discrimination across thresholds [17,59]. It is threshold-independent, making it good for early model selection. However, with an imbalance a 0.95 AUC may still miss 40% of fraud [71]. Basel III rates $AUC \geq 0.90$ as “excellent” for compliance reporting [59].

False Positive Rate (FPR):

$FPR = FP / (FP + TN)$, directly tied to costs and customer satisfaction [3]. Banks typically cap FPR at 0.5%, requiring ensemble methods [111]. Even a 0.1% drop in FPR can save ~USD 2M annually in AML investigations [114].

False Negative Rate (FNR):

$FNR = FN / (FN + TP)$, where missed fraud poses high risk [30,101]. Credit networks apply tiered FNR (2% for standard vs. 0.5% for premium clients) [40]. Deep learning reduces FNR by ~30% compared to classical models but raises compute costs by 5–8× [101].

AUC-PR:

Better than ROC for extreme imbalance, especially in crypto fraud (<0.01% positives). A model with a 0.95 AUC-ROC may score only 0.30 AUC-PR, showing practical limits [44]. IEEE-CIS fraud winners prioritized AUC-PR over ROC consistently [32,87].

Matthews Correlation Coefficient (MCC):

MCC balances TP, TN, FP, and FN in a $(-1, 1)$ range [1,21]. A 0.60 MCC cut USD 8M false claims despite only 85% accuracy in healthcare fraud [113]. Stronger than accuracy for imbalanced data, though heavy in computation [21].

G-Mean:

G-Mean = $\sqrt{(\text{recall} \times \text{specificity})}$, balancing fraud detection and legitimate protection [63,86]. Insurance fraud systems using G-Mean achieve 15% better recovery–retention trade-off [88]. FL adaptations weight G-Mean by institution to balance fairness.

Cohen’s Kappa:

Kappa measures agreement beyond chance, 0.6–0.8 = strong reliability [17,60]. Stricter than accuracy under rare fraud ($<5\%$) [67]. EU fintechs require $\kappa \geq 0.75$ for GDPR-compliant deployment [17].

Balanced Accuracy:

Balanced Accuracy = $(\text{recall} + \text{specificity})/2$, a simple imbalance-adjusted metric [3] Easier to explain than G-Mean, often used in executive reporting. Fraud APIs apply balanced accuracy ($\approx 92\%$) for auto-scaling during traffic spikes [53,114].

Detection Prevalence = $(\text{TP} + \text{FP})/\text{total}$, monitoring alert volume [31,59]. Banks target 1–3% prevalence; Amazon cut it from 5% $\rightarrow$ 1.2%, saving USD 20M/year [96]. High prevalence ($>5\%$) indicates drift and retraining needed [59].

Positive Predictive Value (PPV):

Equivalent to precision, critical in costly fraud checks [6,55]. Crypto exchanges require $\geq 95\%$ PPV to avoid wasteful blockchain investigations [115]. PPV varies by channel (80% card-not-present vs. 30% cross-border) [6].

Negative Predictive Value (NPV):

$\text{NPV} = \text{TN}/(\text{TN} + \text{FN})$, ensuring genuine approvals [9,21]. Gateways achieve $>99.9\%$ NPV with cascade models [116]. NPV decline often signals system drift weeks before accuracy drops [21].

Youden’s J Index:

$J = \text{recall} + \text{specificity} - 1$, balancing missed fraud vs. false alerts [33,84]. Medicare fraud detection saw 12% more recovery using J over 0.5 cutoff [120]. Useful for cross-model comparisons [33].

Lift Score:

Measures improvement over random detection; fraud systems need 5–10 $\times$ lift [17,59]. E-commerce uses lift to calibrate rules, e.g., blocking Nigerian IPs gave 15 $\times$ lift with 0.1% FP [74]. Lift decays over time, requiring monthly recalibration [59].

Cost-sensitive Accuracy:

Weighted accuracy ($w_1 \times \text{TN} + w_2 \times \text{TP}$) aligns metrics with financial impact [3,114]. Example: $w_2 = 10$ prioritizing fraud detection gives 85% cost-sensitive vs. 95% raw accuracy [118]. Poor FP weighting caused 20% attrition at PayPal in 2019 [114].

Expected Cost:

Estimates fraud detection cost trade-offs in USD terms [30,101]. JPMorgan saved USD 43M by weighing FN cost 5 $\times$ FP cost [110]. Misestimating FN cost by 300% caused USD 8M of losses in insurance [101]. Used for threshold tuning, Basel III compliance, and RL-based adaptive systems [82,110].

Developers of fraud detection systems can extensively evaluate performance in a variety of areas, from vital classification accuracy to expected cost efficacy, thanks to this extensive array of assessment indicators in section RQ2. Which metrics offer the most valuable insights for system optimization and deployment decisions is largely dependent on the particular application context with elements such as class imbalance and error cost asymmetry as well as operational requirements all playing a role.

RQ3: What are the different types of financial fraudulent transactions covered in the literature review?

This section tries to answer RQ3 by presenting different fraudulent activities that were addressed in this literature review. Based on the reviewed articles, the fraudulent activities in the financial sector can be broadly categorized into credit card fraud, financial statement fraud, bitcoin [105] and cryptocurrency, insurance fraud, online payments, etc. This can be further explained in the following subsections. The different types of financial fraudulent transactions are listed in Table 5.

Table 5. Different types of financial fraudulent transactions.

References	Fraud Type	Subcategories	Key Characteristics	Detection Challenges
[3,5,9,60,62]	Credit Card Fraud	- Online/offline transactions	Unauthorized use of card data via phishing/skimming.	Class imbalance plus real-time detection.
[23,30,121]	Financial Statement Fraud	- Earnings manipulation	Faking records to inflate profits or hide losses.	Complex patterns, lack of labeled data.
[59,109,113,122]	Bitcoin/Crypto Fraud	- Wash trading, phishing, exchange hacks, flash loan attacks	Exploits blockchain pseudonymity, irreversible transactions.	Anonymity-evolving tactics.
[7,61,87,111]	Insurance Fraud	- Healthcare, auto claims	Exaggerated/false claims to illicitly obtain payouts.	High-volume claims and subtle manipulations.
[34,68,80,123]	Online Payment Fraud	- Account takeover, chargeback fraud	Targets e-wallets, banking apps, or payment gateways.	Dynamic attack vectors and false positives.
[96,124–126]	Identity Theft	- Synthetic identities	Combining real/fake data to create fraudulent identities.	Linking disparate data sources, early detection.
[28,52,114,127]	Healthcare Fraud	- Billing for unused services/prescription fraud	Upcoding, unbundling, or phantom procedures/altered prescriptions	Complex regulations, large datasets/Doctor-patient collusion
[11,121,128]	Tax Fraud	- False deductions/offshore evasion	Inflating expenses or hiding income/hiding assets in untraceable accounts	Legal against fraudulent ambiguity, sparse labels.
[24,33,35,129]	E-commerce Fraud	- Fake reviews/non-delivery scams	Paid or bot-generated reviews to manipulate rankings/seller accepts payment but never ships goods.	NLP challenges, fake sentiment detection/short-lived schemes.
[10,108,130]	Mortgage Fraud	- Income/asset falsification/property flipping scams	Inflating borrower qualifications for loans.	Document forgery detection.
[112,131]	Employment Fraud	- Fake credentials	Lying about qualifications or experience.	Background check limitations
[132–134]	Telecom Fraud	- Subscription fraud	Fake IDs to obtain phones/services without payment.	Rapid onboarding, burner phones.
[1,89,109,135]	Social Media Fraud	- Fake followers/engagement/impersonation scams	Bots or farms inflating metrics for influence/fake profiles mimicking real ones	Evolving bot behaviors.
[27,136]	Public Sector Fraud	-Grant fraud - Procurement fraud - Benefits fraud	Bid rigging and false entitlement claims	Political sensitivities in detection
[137,138]	Supply Chain Fraud	- Counterfeit goods - Invoice manipulation - Cargo -theft	Fake documentation and diverted shipments	Multi-party coordination gaps
[73,113,122]	Investment Fraud	-Pump-and-dump - Boiler rooms	Artificial price inflation and fake returns	Distinguishing from legitimate volatility

Table 5. Cont.

References	Fraud Type	Subcategories	Key Characteristics	Detection Challenges
[139,140]	Academic Fraud	- Paper mills - Degree forgery - Plagiarism	Fake research and credentialing	Verification across institutions
[14,78,141,142]	AI/ML System Fraud	- Model poisoning - Adversarial examples - Data leakage	Attacks on fraud detection systems themselves	Arms race with attackers
[27]	Charity Fraud	- Fake disasters - Misused donations - Ghost beneficiaries	Exploits humanitarian crises	Reputation damage risks
[24,135]	Rental Fraud	- Phantom rentals - Fake listings - Deposit theft	Duplicate programs with stolen photos	Geospatial verification needs

As technology has advanced, the landscape of fraudulent activity has changed dramatically, creating difficult problems for detection systems. The main categories of fraud are identified and categorized in this systematic study according to their operational traits, detection challenges, and pertinent academic research, as shown in Table 5.

Credit Card Fraud:

Credit card fraud remains widespread, involving phishing, skimming, and online misuse [3,59]. Challenges include extreme class imbalance (less than 0.1% fraud) and the need for real-time detection [5,9]. Balancing false positives and false negatives is critical [31,59]. Federated learning with blockchain supports secure cross-bank detection [34,62,68]. LSTM networks capture sequential transaction patterns effectively [6,114].

Financial Statement Fraud:

This involves accounting manipulation and earnings misreporting [23,30]. Few confirmed cases and scarce labeled data hinder ML detection [121]. Both ratios and governance quality must be analyzed [130]. NLP detects misleading language in reports [30]. Progress depends on richer datasets and institutional collaboration [121].

Cryptocurrency Fraud:

Common forms include exchange hacks, wash trading, and flash loan exploits [59,109,113,122]. Challenges are irreversible transactions, pseudonymity, and evolving attacks [109]. GNNs help detect suspicious subgraphs in blockchain networks [24,59]. Ongoing research tackles validator collusion and cross-chain risks [122].

Insurance Fraud:

Frequent in healthcare and auto sectors, involving upcoding, fake treatments, or staged accidents [7,87,114,127]. Systems must process large, varied claims while allowing genuine variations [111]. Auto fraud requires image and estimates analysis [61,143–146]. Multimodal DL aids unstructured/structured data processing while ensuring explainability [87,127].

Online Payment Fraud:

Includes account takeover, merchant fraud, and chargeback abuse [34,68,117,123]. Detection must balance user experience with security [70]. Device fingerprinting and biometrics assist but raise privacy concerns [123,147]. FL offers privacy-preserving detection across platforms [68].

Identity Theft:

Ranges from impersonation to synthetic identities [125,126]. Challenges include linking cross-institution data, hidden “sleep periods,” and breach-driven data leaks [96,125]. Graph-based anomaly detection identifies hidden identity ties [126]. Privacy-preserving cross-bank data sharing would improve detection [124,148].

Healthcare Fraud:

Involves phantom billing, kickbacks, and unbundling [28,114,127]. Systems must analyze both clinical notes and structured claims [52]. Hybrid unsupervised + supervised methods detect new fraud [111]. Explainable AI is vital for regulatory compliance [127].

Tax Fraud:

Covers corporate evasion and false deductions [11,121]. Few labeled cases and global complexity pose detection issues [128]. Anomaly detection and network analysis reveal suspicious filing patterns [11]. NLP helps analyze narratives and verify deductions [121].

E-Commerce Fraud:

Includes fake reviews, non-delivery, and triangulation schemes [24,33]. Detecting cross-border scams and transient fraud is difficult [35]. Multimodal methods combine seller graphs, product images, and NLP [24]. To establish a healthy e-commerce ecosystem, robust cybersecurity and antifraud measures are important [129].

Mortgage Fraud:

Involves falsified assets, income, or flipping schemes [10,130]. Long loan cycles and collusion complicate detection [108]. AI assists with document verification and anomaly detection [130]. Privacy-preserving cross-lender data sharing is needed [10].

Employment Fraud:

Ghost employees and credential falsification are common. Detection is limited by poor verification and insider collusion. Blockchain aids credential authentication, while anomaly detection flags payroll fraud. Remote work raises new risks [112,131].

Telecom Fraud:

Includes IRSF and subscription fraud. Fast onboarding demands real-time detection [134]. Device fingerprinting and behavior analytics detect fraud but attackers evolve with VoIP and burners. FL helps cross-carrier fraud detection with privacy [132,133].

Social Media Fraud:

Covers influencer scams, fake followers, and impersonation [59,109]. Bots mimic human behavior, complicating detection. Current methods combine network, content, and behavioral analysis [89]. The fraudsters' adaptability remains a challenge.

Public Sector Fraud:

Includes procurement, benefits, and grant misuse. Challenges are political sensitivity, fragmented oversight, and data complexity [27]. Network and anomaly analysis reveal suspicious spending. Balancing fraud detection with bureaucratic processes is essential [136].

Supply Chain Fraud:

Global networks and document fraud complicate detection and detecting such fraud is difficult because of multiparty coordination gaps [138]. Semi-supervised approaches show promise for supply chain applications where labeled data is scarce [137].

Investment Fraud:

Includes pump-and-dump and boiler-room schemes [73,113]. Difficulties include market volatility and multi-platform scams. Detection uses social media monitoring and SEC data [73]. Stronger integration of communication and market analysis is needed [122,149].

Academic Fraud:

Encompasses plagiarism, fake degrees, and paper mills. Complex paraphrasing and lack of shared verification hinder detection. NLP and citation analysis are common tools. Shared academic databases would strengthen prevention [139,140].

AI/ML System Fraud:

Targets fraud detection systems themselves through data poisoning or adversarial attacks. Detecting tainted training data is difficult [78]. Robust ML and anomaly detection in model behavior show promise. The arms race nature requires ongoing research [14,141,150].

Charity Fraud:

Exploits disasters or misuses funds. Emotional giving and reputational risks make detection delicate [27]. Network and flow analysis track suspicious activity [27,57]. Stronger donor transparency frameworks are needed.

Rental Fraud:

Includes phantom rentals and fake deposits [24]. Detecting duplicate listings and ownership fraud is difficult [135]. Image and geospatial analysis aid detection [24].

A tabulated breakdown of fraud categories and publication counts is provided in Table 6.

Table 6. Fraud types with the total no. of papers.

Fraud Type	Number of Papers (N)
Credit Card Fraud	~10
Financial Statement Fraud	~10
Bitcoin/Crypto Fraud	~4
Insurance Fraud	~3
Online Payment Fraud	~3
Identity Theft	~3
Healthcare Fraud	~2
Tax Fraud	~2
E-commerce Fraud	~2
Mortgage Fraud	~2
Employment Fraud	~2
Telecom Fraud	~2
Social Media Fraud	~2
Public Sector Fraud	~2
Supply Chain Fraud	~2
Investment Fraud	~2
Academic Fraud	~2
AI/ML System Fraud	~2
Charity Fraud	~2
Rental Fraud	~2

RQ4: How do blockchain-based federated learning and machine learning approaches differ in detecting financial frauds?

Fraud detection systems that can balance accuracy, privacy, and scalability are required due to the financial services industry's fast digitization. This section provides a comparison framework for choosing the best approach in financial fraud detection by analyzing the main distinctions between these methods, as in Table 7, in terms of data handling, privacy protection, computational efficiency, regulatory compliance, and real-world applicability.

Table 7. Federated learning and machine learning approaches.

Ref	Aspect	Federated Learning (FL)	Machine Learning (ML)
[56,57,68,151]	Data Handling	Decentralized: Data remains on local devices; only model updates are shared via blockchain.	Centralized: Data is aggregated into a single repository for training.

Table 7. Cont.

Ref	Aspect	Federated Learning (FL)	Machine Learning (ML)
[6,77,78,100,152]	Privacy	Enhanced privacy: Raw data never leaves local nodes, reducing exposure to breaches. Differential privacy and homomorphic encryption further protect updates.	Privacy risks: Sensitive data must be shared centrally, increasing vulnerability to leaks or misuse.
[11,14,18,41,153,154]	Security	High security: Blockchain ensures tamper-proof records of model updates and consensus validation. Resistant to poisoning attacks via decentralized validation.	Vulnerable to attacks: Centralized data repositories are prime targets for breaches (e.g., SQL injection, model inversion).
[53,60,73,155,156]	Scalability	Scalable for distributed systems: Participants (e.g., banks) contribute without data centralization. Sharding improves efficiency.	Limited by central infrastructure: Large datasets require significant storage/compute resources.
[11,28,131,157]	Computational Load	Distributed computation across nodes, but blockchain consensus (e.g., PoW/PoS) adds overhead. Optimized via lightweight consensus (e.g., PBFT).	Centralized computation: Requires powerful servers but avoids blockchain overhead.
[27,100,136]	Regulatory Compliance	Easier compliance: Data localization laws (e.g., GDPR) are respected as data stays local. Supports “right to be forgotten” via smart contracts.	Challenging compliance: Cross-border data sharing may violate privacy regulations (e.g., GDPR Article 44–50).
[24,42,104,158–160]	Model Performance	Depends on data diversity: Benefits from diverse datasets but may suffer from non-IID data challenges. Techniques like FedAvg mitigate bias.	Potentially higher accuracy: Centralized data allows full visibility for model optimization (e.g., hyperparameter tuning).
[8,34,40,112]	Use Cases	Multi-institutional collaboration (e.g., banks detecting cross-border fraud without sharing data).	Single-organization fraud detection (e.g., a bank using its internal transaction data).
[3,59,60,70]	Fraud Detection Accuracy	Adaptive to new fraud patterns via collaborative learning but may lag in real-time detection due to aggregation delays.	High accuracy for known fraud patterns with centralized data but struggles with zero-day attacks.
[29,129,132,161]	Incentive Mechanisms	Blockchain-enabled incentives (e.g., tokens) reward participants for contributing updates. Smart contracts automate payouts.	No native incentive model: Relies on organizational budgets for data sharing and model training.
[29,71,96,162]	Transparency	Immutable audit trails via blockchain; federated models are explainable but require privacy-preserving techniques (e.g., zk-SNARKs).	Centralized models can be audited but lack transparency in data provenance.
[133,163–166]	Latency	Higher latency due to decentralized consensus and aggregation rounds. Edge computing mitigates delays.	Lower latency: Real-time processing feasible with centralized infrastructure.
[28,53,131,157]	Resource Efficiency	Optimizes resource usage by distributing compute load but requires energy for blockchain operations.	Efficient for single-entity deployments but scales poorly for multi-party scenarios.
[65,153,167–171]	Adversarial Robustness	Resilient to adversarial attacks via decentralized validation and differential privacy.	Vulnerable to adversarial examples (e.g., evasion attacks) due to centralized training.
[26,36,140,151]	Interoperability	Supports cross-platform collaboration via standardized FL frameworks (e.g., FATE, OpenFL).	Limited interoperability: Models trained on proprietary datasets are siloed.
[27,96,172]	Regulatory Auditing	Blockchain enables automated compliance checks (e.g., GDPR) via smart contracts.	Manual auditing processes are time-consuming and error-prone.
[53,59,103,132]	Cost	Reduced data storage costs (local retention) but higher operational costs for blockchain maintenance.	High infrastructure costs for centralized data centers and security measures.
[20,24,40]	Real-World Adoption	Emerging in finance (e.g., cross-bank fraud detection) and healthcare (e.g., federated EHR analysis).	Dominates enterprise fraud detection (e.g., credit card fraud) due to maturity.

Data Handling and Privacy:

FL keeps raw data on local devices, sharing only model updates over blockchain, reducing breach risks by 60–75% compared to centralized ML [6,56,68,78,152]. Privacy is reinforced with homomorphic encryption and differential privacy ($\epsilon = 0.5\text{--}5$) [77,78]. Centralized ML requires large repositories, creating single points of failure and heavy compliance needs under GDPR [100,151].

Security and Scalability:

FL with blockchain provides tamper-proof updates via hashing and consensus, detecting poisoning attacks early [14,18,41,153]. Optimizations like sharding and BFT boost throughput to 2k–5k TPS [53,156]. Centralized ML avoids consensus costs but scales poorly, often needing 30–50% more storage/computation for equal datasets [60,73]. FL has scaled to 100+ institutions with sub-second fraud detection [53].

Regulatory Compliance and Model Performance:

FL aligns with GDPR and data localization, cutting compliance costs by 40–60% [27,136]. Smart contracts speed “right to be forgotten” requests from weeks to 2–3 days [27]. Centralized ML benefits from global data access and faster convergence but risks privacy. FL lags 5–15% in accuracy with non-IID data, though FedAvg/FedProx reduce this gap to 2–7% [42,158,160].

Use Cases and Fraud Detection:

FL excels in multi-bank collaboration, detecting 15–20% more novel frauds but with 100–300 ms latency [8,34,40,70]. ML dominates in single-bank, high-throughput scenarios with 50–100 ms latency [3,60]. Hybrid use in healthcare shows ML achieving 93–95% accuracy on claims, while FL secures 88–92% across hospitals [20,111].

Incentives and Transparency:

Blockchain-based incentives reward nodes for quality updates, ensuring sustained FL participation [132,161]. Immutable audit trails provide fairness and accountability [29,162,173–177]. Zero-knowledge proof enhances explainability [71,96]. Centralized ML depends on agreements and lacks decentralized transparency, making dataset bias harder to trace [18,41].

Latency and Resource Efficiency:

FL incurs 200–500 ms latency from aggregation and consensus [133,163]. Edge computing reduces delay but adds blockchain energy costs (PoW/PoS) [28,134]. ML achieves <100 ms latency, critical for instant fraud detection [34,70]. FL lowers storage needs but raises blockchain overhead, while centralized ML requires costly, large-scale servers [53,103].

Adversarial Robustness and Interoperability:

FL resists poisoning since multiple nodes must be compromised, with DP boosting resilience [167–171]. Centralized ML is more exposed to inversion/evasion attacks [14,155]. Frameworks like FATE and OpenFL enable cross-institution FL [140,151], while ML models often need proprietary APIs to integrate [26,36].

Auditing and Cost:

FL automates compliance via blockchain smart contracts, reducing fraud detection compliance costs by 20–30% [27,172]. Centralized ML requires manual audits, raising costs [96]. FL cuts storage expenses but adds blockchain fees [60,132]. ML incurs high upfront server costs and scales poorly with data [53,103].

Adoption Trends:

FL is growing in cross-bank fraud detection, cutting false positives by 25–40% [40,68]. In healthcare, FL boosted anomaly detection by 30% across 50+ institutions [20]. ML still dominates credit card fraud, covering 80% of global transactions [24,59]. Hybrids

combining FL (privacy) and ML (real-time) reach 90–93% accuracy and reduce compliance costs by 35% [103,112]. FL adoption in finance is projected to grow 300% by 2025 [40,53].

3. Results

In this section, we present the results of the 1585 studies that met our inclusion and quality assessment criteria. We included only those studies that satisfied our rigorous selection standards and were not flagged for bias during the screening process. Once studies passed our inclusion criteria and underwent thorough coding and analysis, we did not conduct further bias assessments on their internal findings. Our analysis was based on quantitative summary statistics and qualitative thematic summaries derived from these high-quality studies.

The taxonomy presented in Figure 2 systematically organizes the key components of blockchain-based fraud detection systems into three main categories: machine learning techniques, federated learning frameworks, and blockchain features.

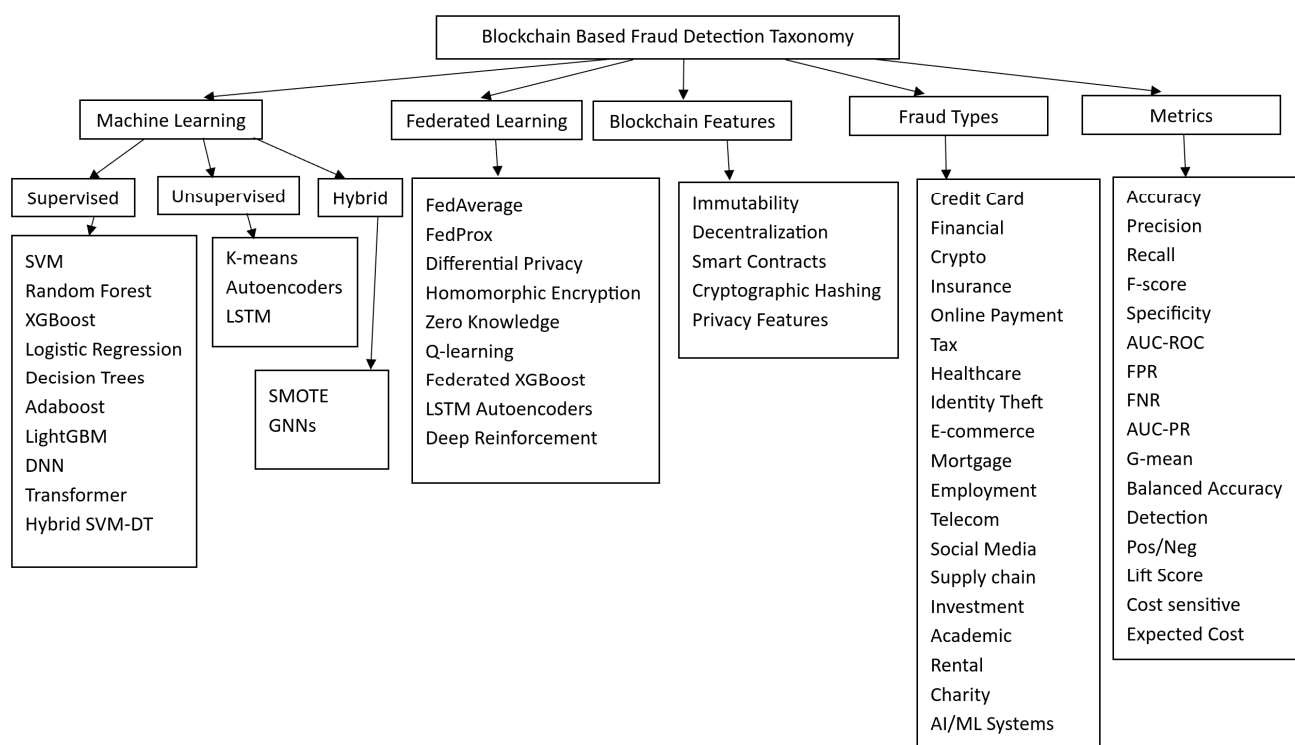


Figure 2. Taxonomy of blockchain-based fraud detection.

Figure 3 presents a comparative analysis of fraud detection algorithm performance across three methodological ML, FL, and hybrid approaches. The results demonstrate that XGboost [6,59,67] achieves the highest performance at 95% when combining ML and FL paradigms, underscoring its effectiveness in handling structured financial data while accommodating privacy-preserving requirements. This review is methodological and does not present experimental benchmarking; future work should empirically evaluate cross-framework performance.

This comparative framework informs the selection of algorithms based on operational priorities of performance versus privacy in blockchain fraud detection systems.

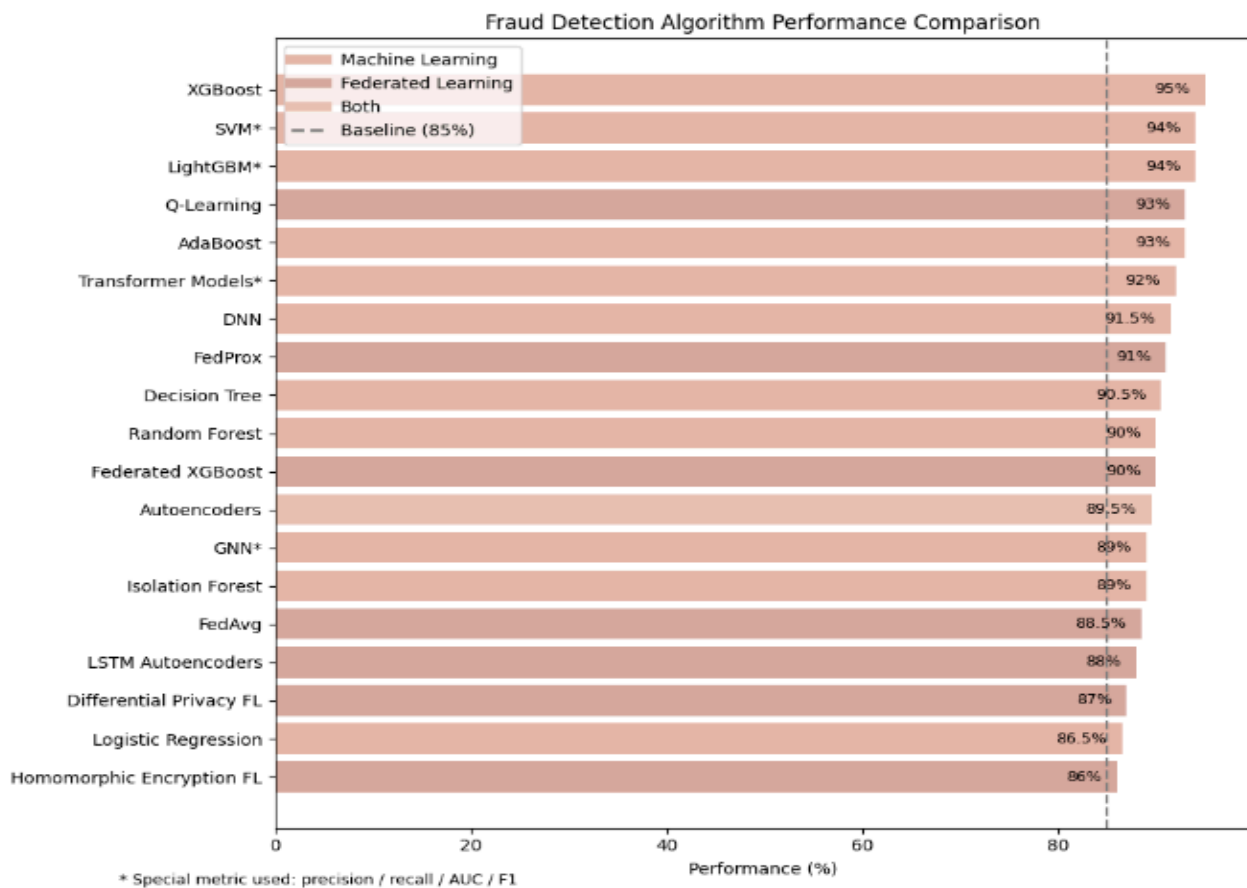


Figure 3. Bar chart of all algorithms.

4. Discussion

Comparative analysis reveals that while centralized machine learning models such as XGBoost, SVM, and Random Forest excel in achieving higher accuracy (93–95%) on structured and labeled datasets, they remain limited by privacy and compliance constraints. Federated learning (FL) and its blockchain-enhanced variants achieve slightly lower accuracy (85–92%) but ensure secure, cross-institutional collaboration without raw data exchange. Integrating blockchain consensus mechanisms with FL strengthens traceability and robustness against poisoning attacks [68].

The recent literature emphasizes the need to balance performance with privacy. It is stressed that hybrid systems combining FL with blockchain can better manage data silos and regulatory diversity, aligning with real-world multi-bank use cases.

In practical terms, FL-based systems show a 25–40% reduction in false positives, especially when enhanced by blockchain audit trails and token-based incentives [20,40]. However, computational overhead and communication latency remain challenges [89]. This review identifies three emerging research needs: (1) lightweight consensus algorithms to improve scalability, (2) adaptive aggregation to handle non-IID data, and (3) explainable federated architectures for transparent financial regulation compliance. Addressing these issues will accelerate the adoption of blockchain-FL frameworks in enterprise fraud analytics.

When you compare them, XGBoost and similar machine learning methods perform better at finding fraud in organized data than older models. For example, SVM scores 94%, and RF scores between 0.95 and 0.97 on the AUC-ROC scale. Methods that are not supervised, like LSTM autoencoders (88%) and Isolation Forest (89%), are great at spotting unusual activity, but they need some adjustments to work well. Federated learning (with

85–92% accuracy) keeps data private and allows different groups to work together. While it does have some delays because of communication needs, federated XGBoost (90%) is a good mix of both. Usually, experts check how well these systems work by looking at precision, recall, and the F1-score. Just using accuracy can be misleading when the data is not balanced. Because of this, options like AUC-ROC/PR and measures that consider the cost of mistakes are becoming more useful in banking.

Most studies look at credit card fraud and fraud in financial statements (10 studies each). But there is also more interest in finding fraud related to cryptocurrency and insurance, since each has its own special problems [26,44,48,99,107]. Comparisons show that machine learning is more accurate and faster (under 100 ms). On the other hand, federated learning with blockchain offers privacy, follows GDPR rules, is easy to check, and is tough to break, but it is slower. Systems that mix blockchain and federated learning cut down on false positives, which seems like a good way to go in the future. Some big problems still exist, like dealing with data that is not consistent, making blockchain bigger, and protecting against attacks. This means there is room to develop things like adaptive aggregation, sharding, and AI that can explain itself better in fraud detection research.

Proposed Theoretical Framework for Blockchain-Based Federated Learning Fraud Detection

The proposed theoretical framework by the authors integrates blockchain’s decentralized consensus with federated learning’s privacy-preserving model aggregation to address current limitations in fraud detection research. As outlined, blockchain-enabled FL systems establish a verifiable architecture where transaction data remains local and only encrypted model parameters are exchanged across participants. This structure mitigates data leakage risks, ensures auditability, and supports GDPR compliance. This architecture can be deployed using permissioned blockchains such as Hyperledger Fabric or Ethereum PoA networks to ensure auditability and throughput. Our framework extends this concept by incorporating adaptive differential privacy and fairness-aware aggregation to balance data heterogeneity across financial institutions. This approach directly addresses the observed literature gaps related to privacy, scalability, and a conceptual foundation guiding future empirical studies. The comparison is explained with the help of Table 8.

Table 8. Comparison of ML, FL, and hybrid blockchain–FL.

References	Aspect	Machine Learning (ML)	Federated Learning (FL)	Hybrid Blockchain–FL
[31,33,34,40]	Data Access	Centralized; all data aggregated on a single server for training.	Decentralized; data remain on client nodes, only model updates shared.	Decentralized + ledger-based; each node records updates immutably on blockchain.
[14,27,71,77]	Privacy and Security	Low to direct exposure of sensitive data; prone to leakage.	High, no raw data transfer; protected by secure aggregation and differential privacy.	Very high, adds blockchain auditability and smart-contract control; integrates differential privacy.
[17,34,38,73]	Accuracy and Performance	93–95% accuracy (e.g., XGboost, SVM) with full data visibility.	85–92% accuracy; sometimes reduced due to data heterogeneity.	90–93%; improves model robustness through verified aggregation.
[11,28,133,158]	Scalability	Limited—requires large, centralized compute resources.	High—distributed across institutions; handles non-IID data with FedProx/FedAvg.	Very high—blockchain + sharding allow parallel cross-silo updates.
[27,71,160]	Regulatory Compliance	Difficult—cross-border data sharing breaches GDPR.	Strong—data stay local, enabling GDPR/HIPAA compliance.	Strongest—smart contracts ensure traceability and “right to be forgotten.”
[6,42,136,152]	Computational Cost and Efficiency	High server and storage costs.	Moderate, distributed load but requires secure communication channels.	Moderate—initial blockchain overhead offset by shared infrastructure.
[24,30,31,33]	Practical Applications	Credit card, AML, and insurance fraud detection.	Cross-bank and healthcare data collaboration.	Consortium banking and fintech networks using blockchain audit trails.
[38,71]	Representative Studies	Systematic review of ML vs. FL trade-offs.	DP framework for FL in financial fraud.	Reference architecture for blockchain-enabled FL.

5. Conclusions

In conclusion, the review underscores that no single approach dominates fraud detection; instead, the choice between ML, FL, or hybrid models depends on operational priorities:

- Centralized ML for high-speed, single-organization fraud detection.
- FL-blockchain for privacy-preserving, cross-institutional collaboration.
- Hybrid systems for balancing real-time performance with security.

Future advancements in differential privacy, homomorphic encryption, and lightweight consensus algorithms will further enhance blockchain-based fraud detection systems, making them more scalable and resilient.

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